MICROCHIP · EMBEDDED SOLUTIONS ENGINEERING · WORKING SPECIFICATION V0.2 · JULY 2026

Garuda SDK

The open autonomy stack for the Microchip drone & robotics platform — perception, navigation and mission services on the Hailo mission computer, paired with PX4 flight control on Microchip silicon over a time-synchronised Ethernet spine.

SCOPE
Companion-computer SDK, three host tiers

LICENCE POLICY
Permissive core — customers ship closed products

PARITY TARGET
ModalAI VOXL SDK

STATUS
Internal draft — figures are planning estimates

Contents

01 Executive summary
02 What Garuda is
03 System architecture
04 Service catalogue
05 State machines
06 The timing chain
07 Drone use cases
08 Competitive comparison
09 Licensing model
10 Roadmap & gates
11 Development plan & AI use
12 Bill of materials
13 Test plan
14 Risks & open items

01 · EXECUTIVE SUMMARY

One SDK, three hosts, and the only open autonomy stack in its power class

Garuda SDK is the software that turns Microchip's mission computer into an autonomy engine — and turns the Hailo acquisition into a platform rather than a part.

- **What it is.** Twelve Linux services — synchronised camera capture, stereo visual-inertial odometry, mapping, planning, NPU inference, the PX4 bridge, time sync, health, logging,

portal and OTA — running on the mission computer and speaking to a PX4 flight controller on PIC32CZ CA90 over Ethernet with PTP.

- **Why it wins.** Every incumbent keeps its VIO closed: ModalAI ships Qualcomm binaries it cannot read; NVIDIA's cuVSLAM is a sealed GEM. Garuda's reference stack is permissively licensed end to end — customers extend it, audit it, and ship closed products on it. The accelerated kernels (v2, Hailo DSP) remain the retained moat.
- **How it de-risks.** Development starts on Raspberry Pi 5 + Hailo-10H (A76 cores remove the CPU question; $130 accelerator; everything host-agnostic), then ports to the Hailo-15H SoM as a BSP exercise with measured per-stage budgets. The programme's biggest unknown becomes a number before the hard money is spent.
- **What it costs.** Under $2,500 of external hardware, two core engineers plus part-time bench with AI-assisted development carrying ~75% of the software authoring, first GPS-denied flight targeted at week 16, usable v1 at month 9.
- **What is genuinely new.** Three pieces: SLAM front-end kernels on the Hailo vector DSP (white space; toolchain access arrives with the acquisition), the PTP-disciplined trigger/strobe camera pipeline as a product, and AI-ISP tuning for night VIO. Everything else is disciplined integration of mature permissive open source — which is the point, not a weakness.

THE ONE-SENTENCE POSITION

ModalAI sells a closed stack on one chip; NVIDIA sells a closed stack on a 15–60 W module; Garuda is the open stack on silicon we own end to end — flight control, propulsion, power, perception, networking and time — and the only one that keeps navigating when the satellites go away.

02 · WHAT GARUDA IS

The companion half of a two-brain aircraft

The platform architecture is deliberately split: hard real-time flight control lives on the PIC32CZ CA90 — its own silicon, RTOS, IMUs and power rail — while everything computationally heavy lives on the mission computer under Linux. Garuda is the mission-computer half. Nothing in the inner control loop ever crosses Ethernet; odometry, obstacle data and mission goals do. If any Garuda service dies, the aircraft is still flown by the flight controller — that separation is a sales argument, an engineering principle and a certification story at once.

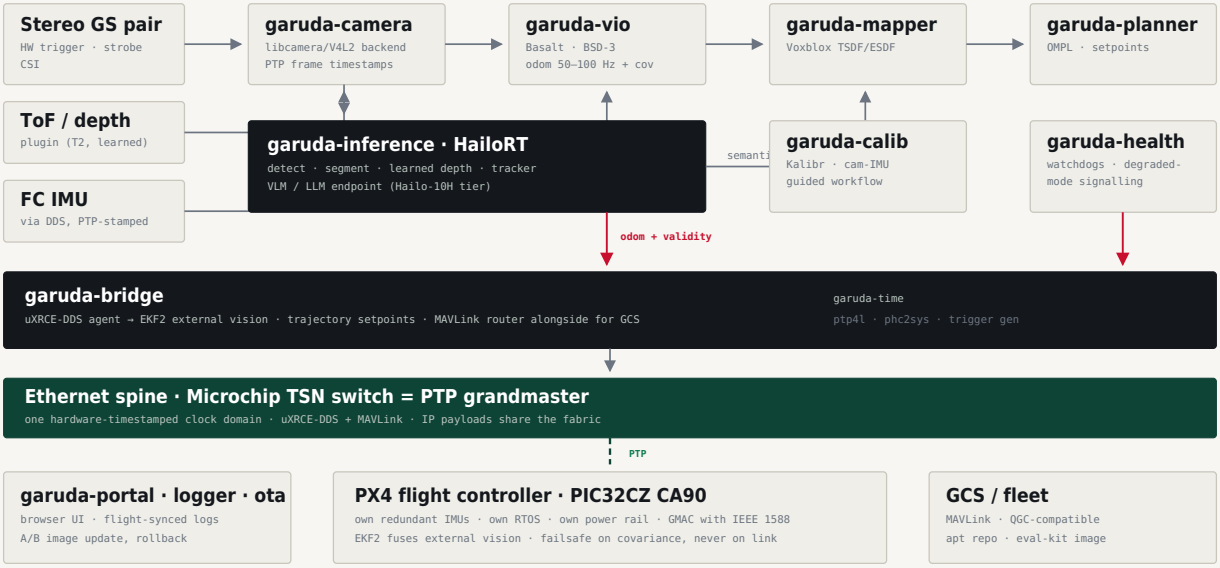
TIER	HOST	ACCELERATOR	ROLE
Dev / eval — now	Raspberry Pi 5 · CM5	Hailo-10H (AI HAT+ 2 bench · bare M.2 flight)	Prove the pipeline, ship the eval kit, on-device VLM demos nobody else can show
Product	Hailo-15H SoM (47×30 mm) on Microchip carrier	integrated NPU + vector DSP + AI-ISP	Industrial tier: −40/+85 °C, sub-0.01 lux night VIO, the two-socket carrier
High end — later	PIC64HX	Hailo accelerator	All-Microchip compute, certification-adjacent missions

DESIGN RULES THE WHOLE SDK OBEYS

1. Failsafe gates on odometry validity and covariance — never on link liveness. Loss of the mission computer is a degraded mode, not a fallout. **2.** One PTP clock domain: camera triggers are generated from the disciplined clock and the strobe returns to a capture input, so the EKF's external-vision delay is measured, not guessed. **3.** The VIO IMU comes from the flight controller over DDS — timestamped at source in the shared timebase. **4.** Every sensor and host sits behind a plugin interface, so tiers and cameras are swappable without touching the estimator.

03 · SYSTEM ARCHITECTURE

Twelve services on one timed fabric



Design rule: attitude and rate control never leave the flight controller. The perception loop (odometry, obstacles, goals) is the only traffic that crosses the spine — and every message in it carries a timestamp from the same disciplined clock, so latency is a measured constant rather than a tuned guess.

FIG. 1 — GARUDA SDK SERVICE ARCHITECTURE AND DATA PLANE

04 · SERVICE CATALOGUE

Twelve services, one contract

SERVICE	WHAT IT DOES (V1.0 FEATURES)	BUILT FROM	LICENCE
garuda-camera	Synchronised capture; hardware trigger out, strobe timestamp in; per-frame PTP timestamps; GS sensors (OV9281 pair via Camarray, IMX296 via XTR); mono and Bayer paths	libcamera / V4L2 + ours	LGPL-dyn + BSD
garuda-vio	Stereo-inertial odometry, 20–30 Hz front end, ≥200 Hz pose via IMU propagation, covariance and validity flags, online time-offset estimation	**Basalt**	BSD-3
garuda-calib	Intrinsics, extrinsics, cam-IMU calibration; guided capture; Allan-variance IMU characterisation	Kalibr + ours	BSD
garuda-inference	HEF model runner (detect / segment / depth / track); model manager; zero-copy frames; VLM endpoint on 10H tier	HailoRT	ours BSD
garuda-mapper	Local TSDF/ESDF obstacle map from stereo or ToF depth; costmap export	Voxblox	BSD
garuda-planner	Local trajectory generation, goal-to-setpoint, stop-and-replan	OMPL, mav_trajectory_generation	BSD / Apache
garuda-bridge	uXRCE-DDS agent; odometry into EKF2 external vision; setpoints back; MAVLink router for GCS	eProsima + PX4	Apache / BSD
garuda-time	ptp4l/phc2sys management; PHC-disciplined camera trigger generation; sync-quality monitoring	linuxptp + ours	GPL daemon + BSD
garuda-health	Service supervision, watchdogs, degraded-mode signalling to the FC	ours	BSD

SERVICE	WHAT IT DOES (V1.0 FEATURES)	BUILT FROM	LICENCE
garuda-logger	Flight-synchronised logging; ULog cross-reference; export tooling	ours	BSD
garuda-portal	Browser UI: status, calibration, configuration, live views, map view	ours	BSD
garuda-ota	A/B image update with trial boot and rollback; apt repository for services	ours	BSD

v1.5 adds: ToF and learned-depth plugins, precision-landing service, night-VIO ISP profile, multi-vehicle time transfer, Rover/AMR robotics profile. v2 (conditional, post-close): DSP-offloaded front-end kernels, NPU-formulated CV, the perception-socket contract as a Dronecode standard proposal. Excluded by policy: GPLv3 estimators (OpenVINS, VINS-Fusion, ORB-SLAM3) and the archived PX4-Avoidance.

05 · STATE MACHINES

The four machines that define behaviour

Autonomy stacks fail at the seams — startup, degradation and update. Garuda specifies those seams as explicit state machines, and the test plan exercises every transition.

5.1 System lifecycle

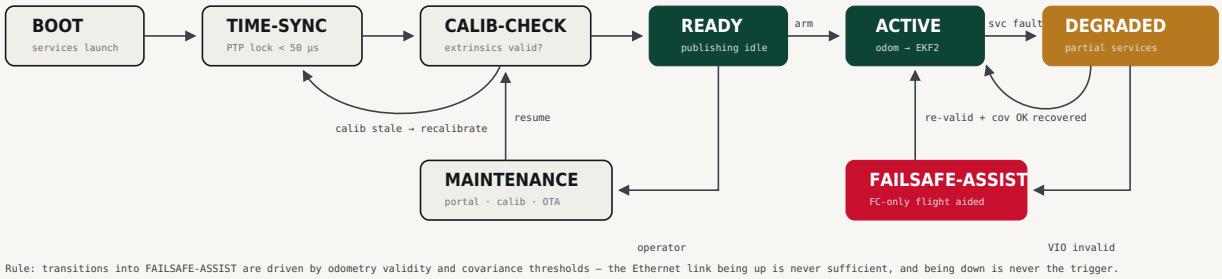


FIG. 2 — SYSTEM LIFECYCLE: PTP LOCK AND CALIBRATION VALIDITY GATE ENTRY INTO SERVICE; DEGRADATION IS A FIRST-CLASS STATE

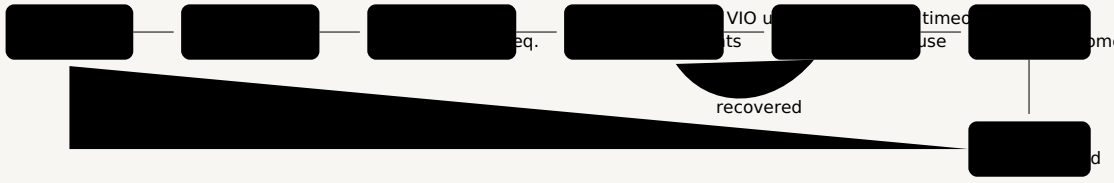
5.2 VIO health



Thresholds τ₁, τ₂ and minimum feature count N are configuration, tuned per airframe during the benchmark programme; every transition is logg

FIG. 3 — VIO HEALTH: COVARIANCE-DRIVEN, WITH AN EXPLICIT ASSISTED RE-INITIALISATION PATH

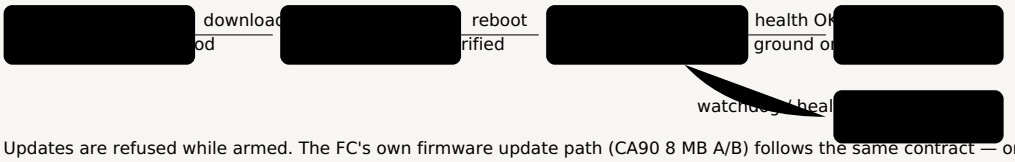
5.3 Mission execution



VIO-LOST during any airborne state hands control to the flight controller's own failsafe (FC-only descent or GPS RTL where available); the missio

FIG. 4 — MISSION EXECUTION: DEGRADATION PAUSES RATHER THAN ABORTS; THE FC ALWAYS OWNS THE LAST WORD

5.4 OTA update



Updates are refused while armed. The FC's own firmware update path (CA90 8 MB A/B) follows the same contract — one story across the platfor

FIG. 5 — OTA: TRIAL BOOT WITH AUTOMATIC ROLLBACK; NEVER IN THE AIR

06 · THE TIMING CHAIN

Why the clock is the product

Every VIO stack fights the same enemy: unknown latency between what the camera saw and what the IMU felt. Garuda's answer is structural — one disciplined clock owns the trigger, the strobe, the IMU and the estimator.

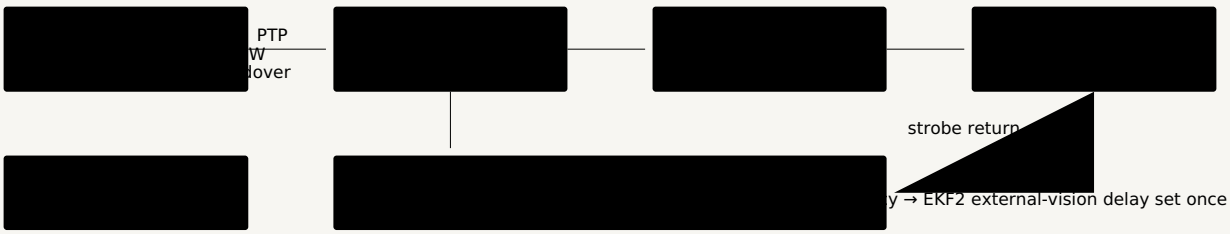


FIG. 6 — THE TRIGGER/STROBE LOOP TURNS EKF2'S DELAY PARAMETER FROM A TUNING EXERCISE INTO A ONE-TIME MEASUREMENT

WHY COMPETITORS CANNOT COPY THIS CHEAPLY

The chain only closes because the switch, the PHYs, the oscillators and the holdover clocks are catalogue parts of the same vendor building the SDK. ModalAI cannot buy an atomic-holdover story from Qualcomm at any price; a DIY integrator can assemble the pieces but inherits the qualification burden Garuda ships as a done deliverable.

07 · DRONE USE CASES

Six missions, one stack

Each use case below names the services that carry it and the Microchip silicon it pulls through — because the SDK exists to sell the platform, and every mission should read as a bill of materials.

GPS-denied industrial inspection

Under bridges, inside plants, beside structures that mirror and mask GNSS. VIO carries navigation; the mapper keeps standoff distance; detection flags corrosion or intrusion on the NPU.

`vio · mapper · planner · inference · bridge`

Pulls through: CA90 FC, Hailo, TSN switch, 6S ESC/power.

Night operations

Hailo-15's AI-ISP holds VIO feature stability below 0.01 lux — navigation in darkness, not just imaging. v1.5's night profile tunes the ISP for photometric consistency rather than pretty pictures. Almost no incumbent answer exists.

`camera (night profile) · vio · inference`

Pulls through: Hailo-15 tier exclusively — a reason to move up from the Pi tier.

Precision landing & delivery

The last four metres: down-camera marker detection fused with ToF depth, immune to low-texture ground that defeats stereo, active illumination for night pads.

`inference (marker) · depth plugin (T2/learned) · planner · bridge`

Pulls through: the depth-plugin contract — any customer sensor drops in.

BVLOS corridor logistics

Long routes through urban canyon and tunnels where GNSS ride-through decides certification credibility. ZL30xxx holdover keeps the clock honest through outages; health/failsafe states give the regulator a defensible degradation story.

`time (holdover tier) · vio · health · logger (evidence)`

Pulls through: ZL30xxx, CA90 8 MB A/B OTA, telemetry + Remote ID module.

FPV / interceptor perception

Target acquisition under 10 ms and 30+ FPS tracking — Hailo's own demonstrated workload — grafted onto the 30.5 mm PX4 FC tier. Garuda's tracker service feeds intercept guidance over the same bridge contract.

`inference (tracker) · bridge · time`

Pulls through: CA70/CA90 FPV FC, 12S ESC, analog VTX path.

Multi-vehicle & ground robots

A shared PTP domain gives swarms a common timebase for sensor fusion and coordinated capture. And the identical stack drives AMRs and rovers — the robotics profile swaps the vehicle model, not the services. Physical AI, one SDK.

`time (multi-vehicle) · full stack, Rover profile`

Pulls through: the §17 robotics adjacency — same silicon, ground variant.

08 · COMPETITIVE COMPARISON

Against the autonomy stacks that exist

	GARUDA SDK (MICROCHIP + HAILO)	VOXL SDK (MODALAI / QUALCOMM)	ISAAC ROS (NVIDIA JETSON)	AUTERIONOS / SKYNODE	DIY PX4
VIO core	**Basalt, BSD — source shipped**	**mvVISLAM — closed Qualcomm 32-bit binaries**	**cuVSLAM — closed GEM**	integrates third-party	GPL typi
Customer ships closed product	**Yes — permissive end to end**	Partly (glue open, core closed)	Partly (wrappers open)	Commercial licence	**GPL cont risk**
Host power class	**2.5–7 W (10H / 15H tiers)**	~6–10 W	**15–60 W**	~10–15 W	vari
Timing / GNSS-denied clock	**PTP chain + holdover to atomic — unanswerable**	**GNSS-derived, no holdover**	**none productised**	**none productised**	asse your
Fault isolation	**Separate FC die, rail, RTOS**	Partition on one die	Companion-only (needs FC)	**Separate FMU on module**	depe
Night capability	**AI-ISP < 0.01 lux (15H tier)**	Conventional ISP	Sensor-dependent	Sensor-dependent	Sens depe
Maturity / field hours	**New — v1 at month 9**	**Shipping since 2019, Blue UAS**	**Large deployed base**	**Enterprise deployments**	vari
On-device VLM / LLM	**Hailo-10H tier, 3 W — unique demo**	No	Yes at 25–60 W	No	No
Silicon cost (compute)	**~$60–80 class**	Module low hundreds; board $1,199	**$249–$1,999 modules**	Four figures	vari
Vendor scope	**Whole aircraft: FC + ESC + power + comms + timing + perception**	One SoC + boards	Compute module only	FC + OS (buys compute)	none

	GARUDA SDK (MICROCHIP + HAILO)	VOXL SDK (MODALAI / QUALCOMM)	ISAAC ROS (NVIDIA JETSON)	AUTERIONOS / SKYNODE	DIY PX4
Business model	Free SDK sells owned silicon	Hardware margin + Blue UAS position	Free SDK sells $$$ modules	Software licence + hardware	engi

Reading it honestly

Where they win

Maturity, everywhere — VOXL has six years of flight hours and Blue UAS; Isaac has the research mindshare and raw compute; Auterion has enterprise fleet tooling. Garuda's answer is not to claim maturity it lacks but to publish benchmarks, logs and losing rows until the evidence accumulates.

Where Garuda wins now

Openness of the core, licence cleanliness, power class, silicon cost, fault isolation, night capability, the timing chain, and one-vendor scope. Five of those eight are structural — they follow from owning the silicon — and cannot be answered by a software release from any competitor.

The strategic difference

Every rival monetises the stack or the module. Garuda is free and open because it exists to sell fifteen other Microchip parts on the same aircraft. That inverts the incentive: the more customers fork it, the more silicon ships — which is why openness is the strategy, not a concession.

09 · LICENSING MODEL

Permissive by policy, enforced by CI

LAYER	LICENCE	RATIONALE
Garuda services (ours)	BSD-3-Clause	Fork-friendly; every fork is a design-in
Inherited estimator / mapping / planning	Basalt BSD-3 · Voxblox BSD · OMPL BSD · Nav2 Apache	Customers ship closed products legally
Bridges & middleware	uXRCE-DDS Apache · MAVLink LGPL · ROS 2 Apache	Ecosystem standard
linuxptp		

LAYER	LICENCE	RATIONALE
	GPL — daemon at process boundary	No linking, no contamination
HailoRT / DFC models	Hailo terms (open core; compiler gated)	Internal post-close
v2 DSP kernels	Closed firmware — the retained moat	Same pattern as NVIDIA/ Qualcomm, inverted: glue open, kernels ours
Excluded by policy	GPLv3 estimators (OpenVINS, VINS-Fusion, ORB-SLAM3) · archived PX4-Avoidance · CI licence-scan fails the build on ingress	

10 · ROADMAP & GATES

v1 parity → v1.5 differentiation → v2 moat

RELEASE	CONTENTS	GATE TO PROCEED	TARGET
v1.0	Twelve services, VOXL feature parity, eval-kit image, apt repo	VIO ≥20 Hz front end · ≥200 Hz pose · <20 ms latency on Pi 5; PTP <50 µs RMS; camera jitter <100 µs; hover drift <0.5 m / 5 min	Month 9
v1.5	ToF + learned depth plugins, precision landing, night-VIO profile, multi-vehicle time, AMR profile	v1 shipped; two external design-ins active	Month 12–15
Hailo-15 port	Yocto layer, capture backend, HEF recompile, parity report	Delta table vs Pi baseline → sizes or cancels v2	Month 12
v2.0	DSP front-end kernels, NPU CV, Dronecode socket standard, patents	Measured A53 deficit; Cadence toolchain access verified post-close	Month 18+ conditional

11 · DEVELOPMENT PLAN & AI-ASSISTED MODEL

Sixteen weeks to flight, nine months to v1

PHASE	WEEKS	DELIVERS	GATE	AI SHARE
0 Scaffold	0–2	Repo, interfaces, harnesses, install scripts (fixes pre-baked), bridge configs both sides	CI green before hardware arrives	85%
1 NPU	2–3	Pi 5 + AI HAT+ 2 running models + VLM	YOLO ≥30 fps	90%
2 Stereo	3–6	Camarray synced; IMX296 XTR path (1.8 V, R11)	jitter <100 µs, 1 h no drops	70%
3 Time	4–7	PTP to LAN966x GM; disciplined trigger; strobe loop	<50 µs RMS switched	75%
4 VIO	6–10	Calibrated Basalt live; profiler per stage	**the go/no-go numbers**	80% code / 0% number
5 Bridge	8–12	EKF2 fusion, indoor position hold, covariance failsafe demo	drift <0.5 m / 5 min · CA90 Ethernet verified **week 1**	85%
6 Load + CM5	10–14	Concurrent NPU+VIO soak; CM5 + bare M.2 port	no VIO deficit, no throttle, no brownout	60%
7 Flight	12–16	Outdoor GPS-denied hover + waypoints, published logs	the demo asset	50%
8 Productise	m4–9	Portal, logger, OTA, docs, licence audit, eval image	v1.0	80%
9 Hailo-15 port	m6–12	Yocto, backend swap, parity report	delta table → v2 decision	65%

THE AI-ASSISTED WORKING RULES

Artifacts before code — real `uname -a`, package versions, camera SKUs and CA90 driver state are fed in before configs are generated. **Repo, not fragments** — output lands as reviewable commits with tests. **Numbers are sacred** — no benchmark is ever invented; gates come off the bench into `datasets/baselines/`. **Structural review at midpoints. Red-team each phase** — every phase closes with an explicit "argue this result is wrong" pass. Net effect measured on prior ports: ~75% of software authored by AI, calendar compressed by smaller team rather than shorter tail — the bench debug time remains human and irreducible.

Team: 2 core engineers + part-time hardware/bench through Phase 7; +1 for productisation; the same pair executes the Hailo-15 port. Humans own: all wiring and scope work, every empirical number, flight ops, ISP tuning judgement, architecture sign-off, and the v2 Cadence-toolchain kernels.

12 · BILL OF MATERIALS

Under $2,500 external for the whole programme

Bench (order now) ≈ $1,300

Raspberry Pi 5 8 GB ×2 + Active Coolers + 27 W PSUs	$194
AI HAT+ 2 (Hailo-10H 40 TOPS, 8 GB) ×2	$260
Arducam OV9281 stereo + Camarray HAT	~$120
Raspberry Pi GS (IMX296) ×2 + lenses	~$150
Arducam T2 ToF kit	$300
USB camera, on-board H.264 (recording)	~$60
IMU breakout, storage, cables, misc	~$180
LAN9668-class switch EVB · CA90 FC board	internal

Flight prototype ≈ $900

CM5 8 GB/64 GB (no-wireless) ×2	$390
CM5 IO board + compact dual-CSI/ M.2 carrier ×2 (verify Hailo compat — one carrier family known-bad)	~$145
Bare Hailo-10H M.2 2242 8 GB ×2 (order week 1 — sourcing friction)	~$390
5 V ≥6 A vehicle regulator	~$30
Test airframe + your ESC/power boards	internal

Plus one VOXL 2 ($1,199) for the comparative benchmark — the same philosophy as the ESC dyno: beat "good enough" with published measurements, including the rows we lose.

13 · TEST PLAN

Seven layers, every gate a number

LAYER	WHAT RUNS	PASS CRITERIA
CI	Unit + contract tests, x86 & cross builds, static analysis, **licence scan**	Green; build fails on GPLv3 ingress
Dataset regression (nightly)	EuRoC MH/V + TUM-VI through garuda-vio	No metric worsens >5% vs stored baseline

LAYER	WHAT RUNS	PASS CRITERIA
	headless; ATE/RTE, Hz, per-stage ms	
Sync rig	LED flash + strobe capture across both cameras	Inter-camera + trigger→exposure jitter <100 µs
Time rig	24 h PTP offset vs grandmaster, switched + point-to-point; Allan deviation	<50 µs RMS switched; sub-µs P2P sanity
Load rig	VIO + detection + VLM concurrent; thermals, 5 V rail scope, 1 h soak	No VIO frame deficit, no throttling, no brownout
Fault injection	Kill each service mid-run; camera unplug; PTP loss; VIO kill mid-hover	Health signals FC; failsafe fires on covariance, never on link
Flight cards	Hover 5 min · square 20 m legs · VIO return-to-start · night (15H tier)	Drift <0.5 m/5 min; ULog + Garuda log bundle published internally

Phase 9 re-runs layers 2–6 on the Hailo-15H; the delta table is the sole input to the v2 DSP decision. Comparative flight cards on the purchased VOXL 2 feed the pitch deck.

14 · RISKS & OPEN ITEMS

Named, owned, front-loaded

RISK	MITIGATION
CA90 PX4 Ethernet/PTP driver immaturity	Verify in week 1 — the flagged schedule risk; Phase 5 depends on it
Basalt throughput on A53 (port)	Pi-first converts unknown → measured deficit; DSP offload is the sized response, not a bet
Bare 10H M.2 sourcing	Order week 1; AI HAT+ 2 covers the bench regardless
CM5 carrier Hailo compatibility	Buy two candidates; one family already reported incompatible
Cadence Xtensa licence transfer	Pre-close question list — linchpin of v2; verify before any kernel R&D
Hailo-15 mono-GS ISP support	Pre-close technical question; fallback = ISP bypass on the VIO path
GPL ingress by habit	CI licence gate + audit doc

RISK	MITIGATION
Scope creep	v1 list frozen by this document; additions queue for v1.5
Temperature expectations	CM5 is −20/+85 °C (no −40 variant exists); AI HAT+ 2 is 0–50 °C bench-only — defence cold-soak lives on the Hailo-15 tier

THE CLOSING FRAME

ModalAI needed six years because they also built the real-time partition, the IO layer and the VIO from Qualcomm's closed parts. Garuda inherits the NPU runtime with the acquisition, already flies the real-time half on our own silicon, and assembles the middle from mature permissive open source. Same destination — a fraction of the distance, and the only stack in the class that arrives open.

Document generated July 2026. All prices, dates and effort figures are planning estimates pending real quotations and bench measurements. Companion documents: garuda-sdk-plan.md (working plan), the platform strategy document, and the two deep-research reports (Garuda SDK parity analysis; Pi 5 + Hailo-10H development plan).